



Foretellix Highway and ADAS Traffic Scenario Library

Scenario Definitions at the Functional Level

Version 1.0

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1. About this library

The [SAFE](#) Autonomous Vehicle Initiative, in collaboration with Foretellix Ltd, is proud to participate in the UNECE/WP.29/GRVA effort to establish a scenario catalogue aimed at validating the safety of automated driving systems.

Foretellix is offering this set of functional scenarios to be included in the scenario catalogue being developed by WP.29/GRVA/VMAD.

Foretellix will be happy to receive any feedback with respect to these scenarios. When more scenarios are developed, they may be contributed as well.

These scenarios are described in a free-form natural language, matching the “functional scenario” definition of natural language description without any numerical ranges (Pegasus Terminology).

In addition to these functional definitions, Foretellix has created implementations of these scenarios with parameter ranges and coverage measurements for usage in virtual testing. Foretellix’s scenarios are implemented using M-SDL – a scenario description language that will be fully compatible with ASAM OpenSCENARIO 2.0, once it is released. These implementations are the “logical” abstraction level. This level of abstraction is currently outside the scope of VMAD’s New Assessment and Testing Methods document.

The scenario descriptions use the term *DUT* to indicate the Device Under Test, which can also be referred to as the ego car, or as SUT (System Under Test).

2. Cut in

2.1. Scenario description

The Cut-in scenario is a special case of lane change in which a car traveling in the same direction as the DUT pulls into the lane ahead of the DUT. Depending on the cars' speeds and distances (but also on other factors like visibility, road topology), the maneuver leads to accidents, such as collisions.

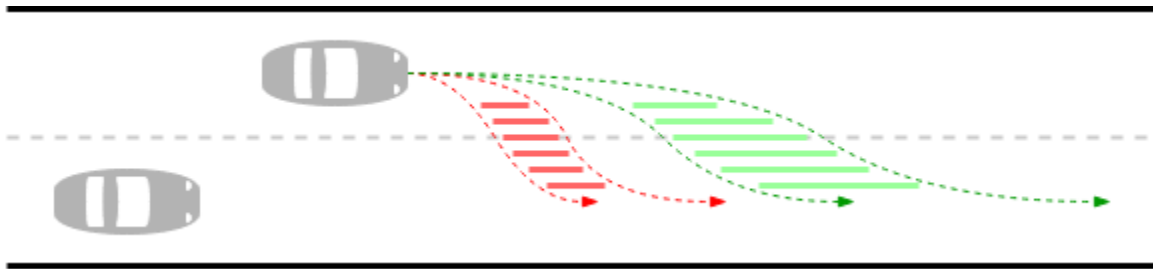


Figure 2-1 Dangerous and less dangerous cut-in maneuvers

Environmental requirements: A road with at least two lanes.

DUT behavior: The DUT keeps the same lane and presumably moves at a constant speed.

Other actors' behavior: At the start of the scenario, another car is in a neighboring lane, ahead of the DUT. At the end, the other car has changed into the DUT's lane and is in front of the DUT.

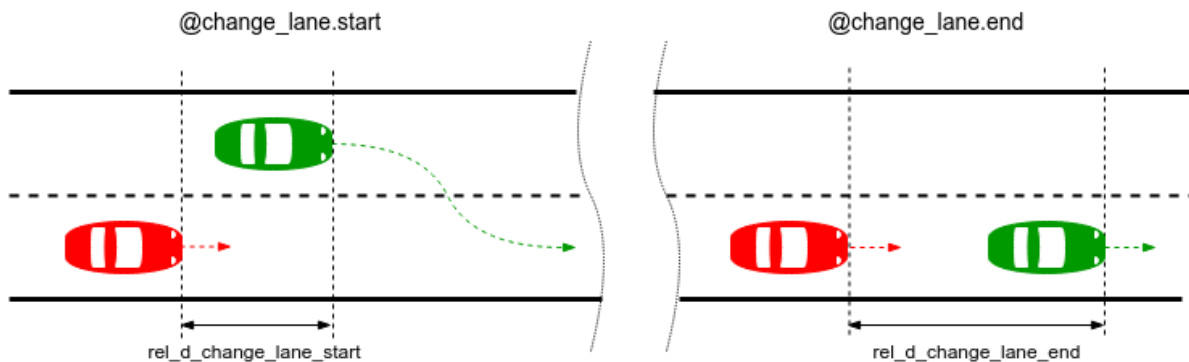


Figure 2-2 Cut-in scenario

3. Interceptor

3.1. Scenario description

For the DUT, junctions present a challenge due to the increased likelihood of conflicts with other actors.

In this scenario, the DUT traverses an intersection simultaneously with another car - the interceptor. This scenario tests the DUT's behavior when on a collision course with another car in an intersection, possibly with signs, signals, or traffic lights. The DUT should be able to safely maneuver through the intersection and avoid or mitigate a collision.

Environmental requirements: A junction with at least three ways. It may or may not be controlled (i.e. have yield sign, traffic lights, etc).

DUT behavior: The DUT traverses the junction in any direction (left, right or straight).

Other actors' behavior: Another car approaches the same junction, from a different direction and traverses the junction such that its trajectory intersects with the DUT's trajectory.

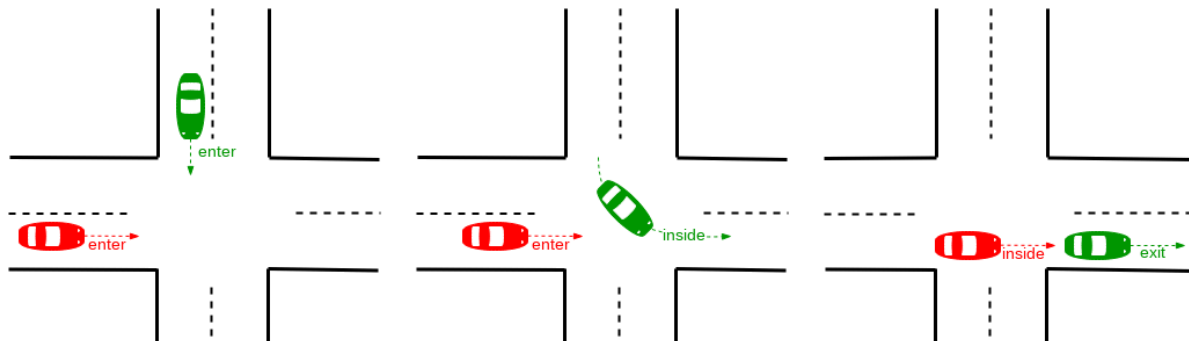


Figure 3-1 Interceptor scenario

4. Highway Scenarios

4.1. Scenarios description

Highways pose a different challenge for the DUT traversing a junction: denser traffic, potentially moving at high speed.

4.1.1. DUT merges at highway entry

Environmental requirements: A highway junction with an entrance/on ramp.

DUT behavior: The DUT performs a merge entering the highway from the entrance/on ramp, possibly at high speed.

Other actors' behavior: Another car drives on the highway approaching the same junction and traverses such that its trajectory intersects with the DUT's trajectory, i.e. exiting the highway into the DUT's merging lane.

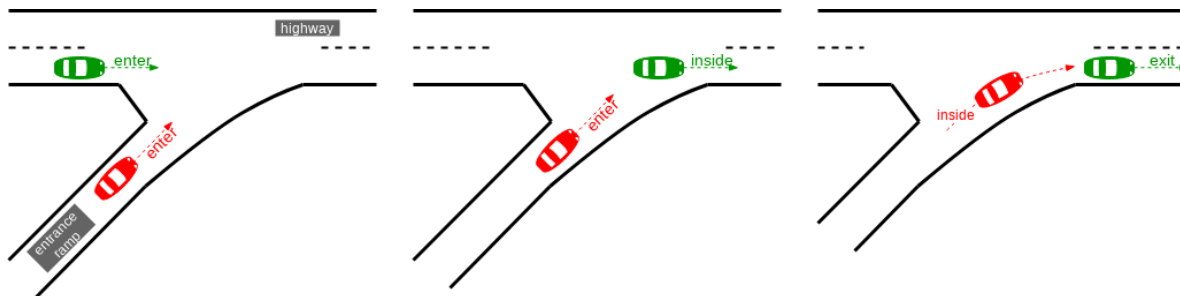


Figure 4-1 DUT merge at highway entry scenario

4.1.2. Cut-out to highway exit

Environmental requirements: A highway junction with an exit/off ramp.

DUT behavior: The DUT drives straight on the highway in the innermost lane.

Other actors' behavior: Another car starts in front of the DUT as lead vehicle from the DUT's perspective. Then it cuts out and moves to an exit/off ramp.

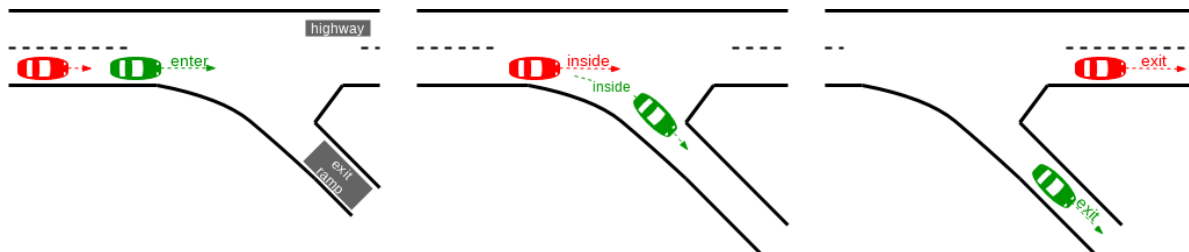


Figure 4-2 Cut-out to highway exit scenario

4.1.3. Cars merge at highway entry

Environmental requirements: A highway junction with an entrance/on ramp.

DUT behavior: The DUT drives straight on the highway in the innermost lane.

Other actors' behavior: Multiple cars perform a merge entering the highway from entrance/on ramp. The time gaps between cars permits the DUT to align within the convoy.

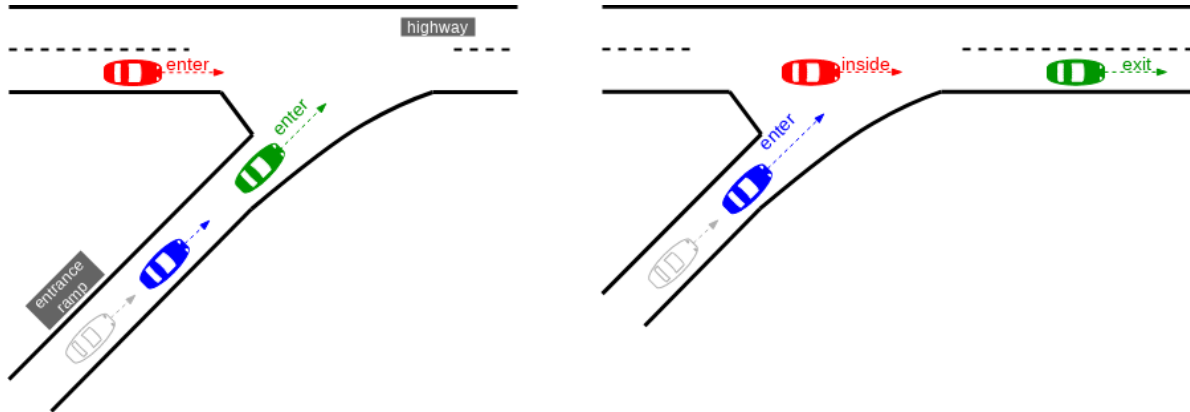


Figure 4-3 Cars merge at highway entry scenario

5. Cut out

5.1. Scenario description

In this scenario, the DUT follows a car that suddenly changes lanes, possibly exposing an obstacle ahead such as a stationary car. Also, if the DUT is part of a platoon, a car cutting in and out of the platoon's lane breaks its coherence. The platoon must automatically adjust the distance to the desired platooning distance. If the gap closes too slow, the cruise velocity may be adapted to ensure that the gap is closed in a limited time.

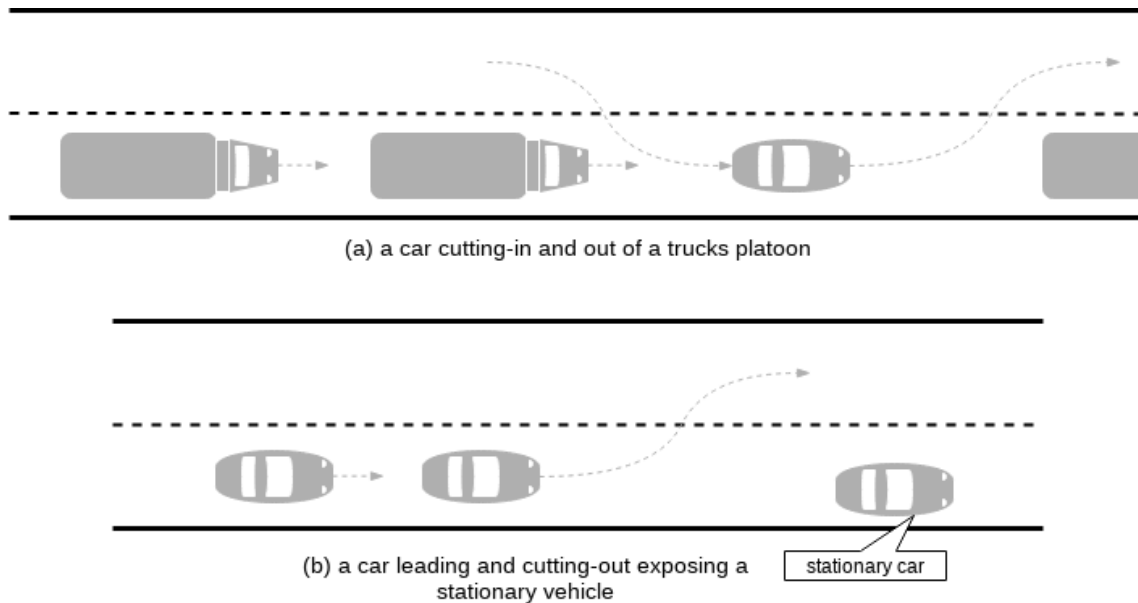


Figure 5-1 Cut-out maneuvers

Environmental requirements: A road with at least two lanes.

DUT behavior: The DUT drives in a lane, presumably at a constant speed.

Other actors' behavior: At the start of the scenario, another car is in the same lane, ahead of the DUT. At the end, the other car has performed a lane change into an adjacent lane.

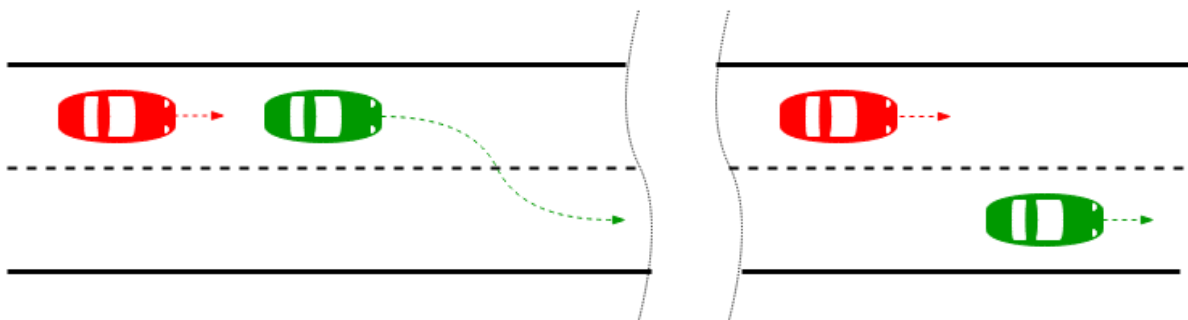


Figure 5-2 Cut-out scenario

6. Oncoming

6.1. Scenario description

Oncoming is a scenario in which a car approaches the DUT from the opposite direction and drives past the DUT.

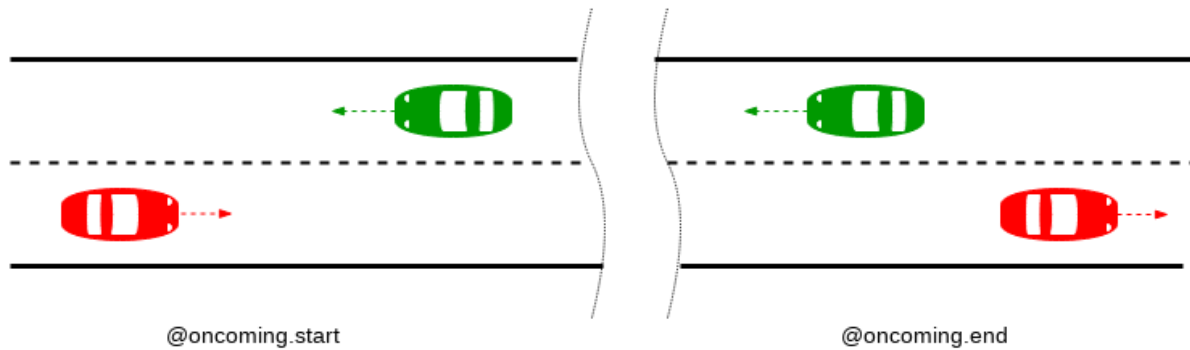


Figure 6-1 Oncoming scenario

Environmental requirements: A two-lane road with traffic moving in opposite directions.

DUT behavior: The DUT drives in a lane, presumably at a constant speed.

Other actors' behavior: At the start of the scenario, another car is in the opposing lane, approaching the DUT. At the end, the other car is still in the opposing lane, having passed the DUT.

7. Slow Neighbors

7.1. Scenario description

Slow neighbors is a scenario in which the DUT passes by a convoy of cars moving at a low speed in an adjacent lane.



Figure 7-1 Slow neighbors scenario

Environmental requirements: A road with at least two lanes.

DUT behavior: The DUT drives in one lane.

Other actors' behavior: The other cars drive in an adjacent lane at a low speed.

8. Lane Hugger

8.1. Scenario description

Driving squarely in the middle of the lane is a challenge for every new human driver, and on roads without shoulders or with parked cars, even experienced drivers may drive close to the adjacent lane. A car with such behavior - driving close to a lane marking - reduces the distance the DUT requires to perceive a potential cut-in maneuver.

Environmental requirements: A road with at least two lanes.

DUT behavior: The DUT drives in a lane, presumably at a constant speed.

Other actors' behavior: Another vehicle starts in a lane adjacent to the DUT's lane. The other vehicle pulls in close to the DUT's lane.

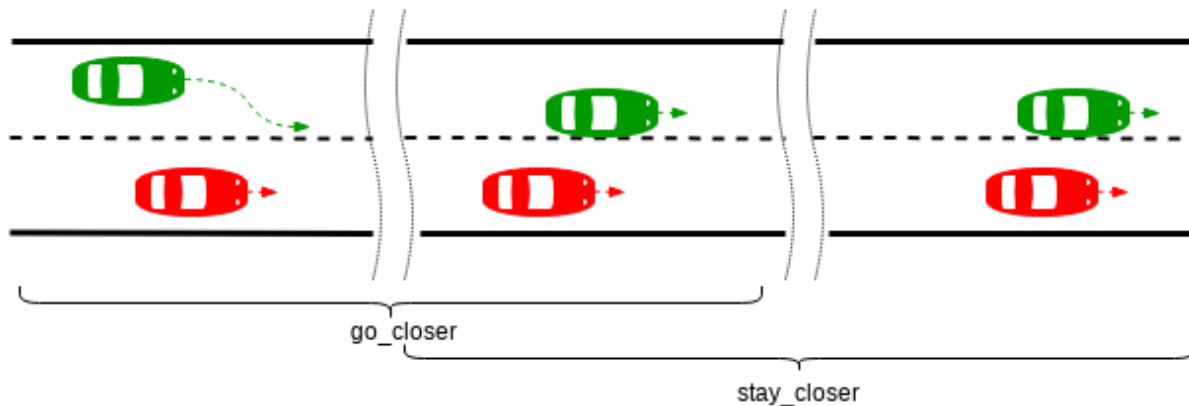


Figure 8-1 Lane hugger scenario

9. To Side on Curve

9.1. Scenario description

Driving on a straight road mainly requires longitudinal control of the vehicle, with some lateral correction to stay in lane. Driving along a curve requires the DUT to determine a speed appropriate for the road curvature and to follow the curve while maintaining lane position. In general, there is a distinction between accidents caused by leaving the road on a straightaway and accidents caused by leaving the road on a curve. Where the former is a failure of lateral control to keep the car in the lane, the latter is usually due to the speed being too high for the road curvature. Either the curvature has not been anticipated correctly, or the speed has not been adapted sufficiently.

Environmental requirements: A curved road with at least two lanes.

DUT behavior: The DUT drives in a lane, presumably at constant speed.

Other actors' behavior: Another vehicle starts in a lane adjacent to the DUT's lane, and drives along the curve ahead of the DUT.

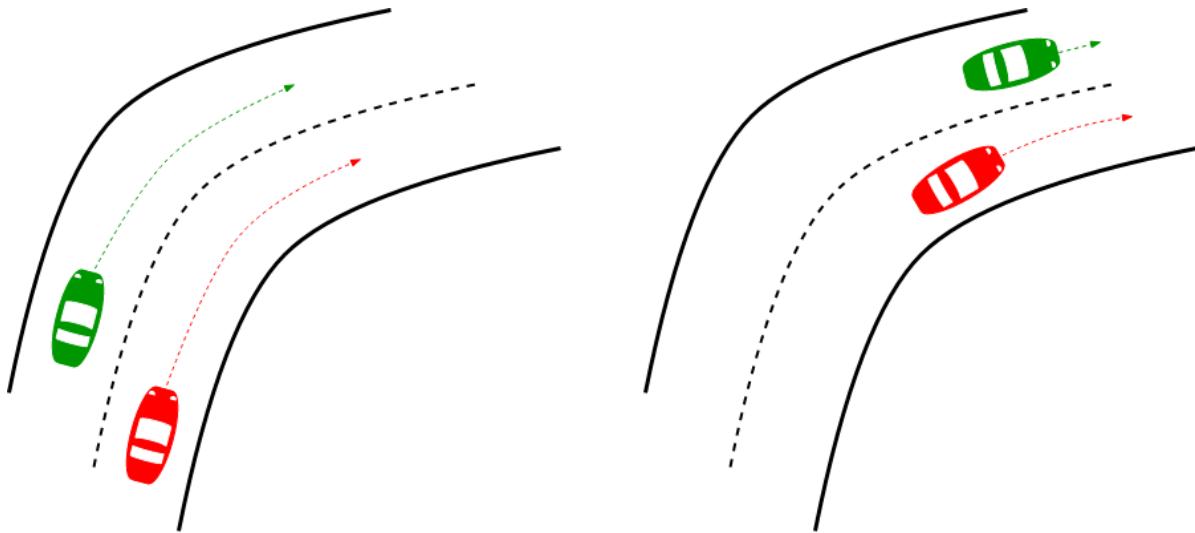


Figure 9-1 To side on curve scenario

10. Stopped Vehicles

10.1. Scenario description

The DUT drives in the vicinity of a convoy of stationary cars.

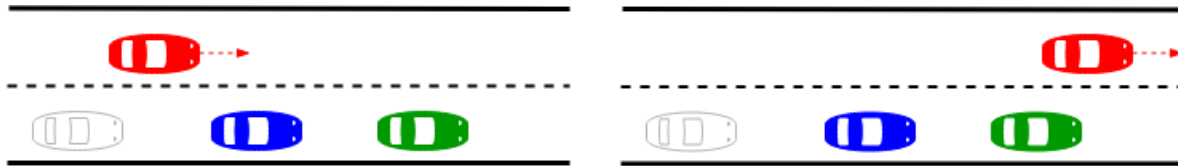


Figure 10-1 Stopped vehicles scenario

Environmental requirements: A road with at least two lanes.

DUT behavior: The DUT drives in one lane.

Other actors' behavior: The other cars are stopped either in the same lane or in an adjacent lane to the DUT.

11. Lead Vehicle

11.1. Scenario description

The purpose of the lead vehicle family of scenarios is to verify the DUT's ability to drive behind another leading vehicle. There are different variations of the base scenario intended to impose different road or traffic situations in which the following of the lead vehicle is performed.

Environmental requirements: A road with a length of at least 100 meters. In some variants (such as lead_vehicle_cut_out) the road needs to have at least two lanes.

DUT behavior: The DUT drives in a lane behind the lead vehicle.

Other actors' behavior: At the start of the scenario, another car is in front of the DUT in the same lane. The other car continues driving in the same lane, in front of the DUT throughout the scenario.



Figure 11-1 Lead vehicle scenario

12. DUT speed limit change

12.1. Scenario description

This scenario challenges the DUT to respond appropriately to speed limit changes by decelerating when entering a lower speed zone and accelerating when entering a higher speed zone. In the example shown below, the speed limit decreases from 80kph to 60kph.

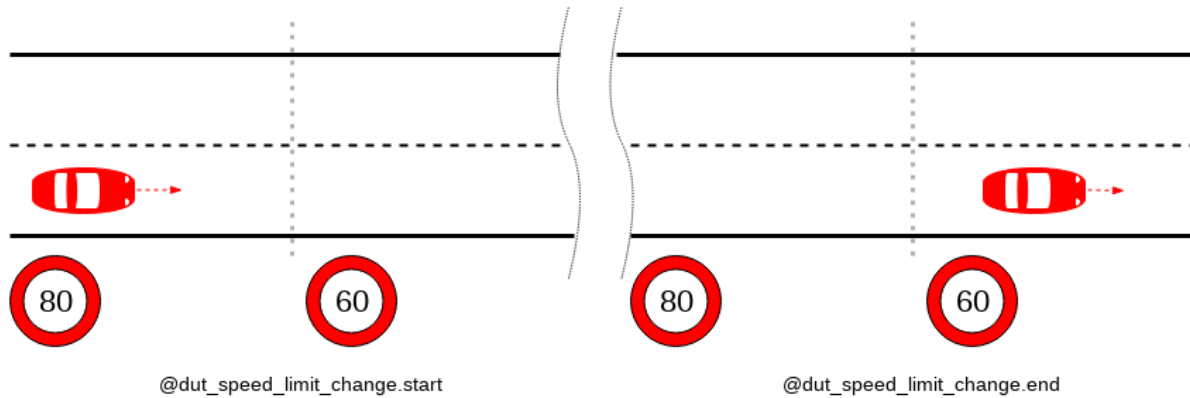


Figure 12-1 dut_speed_limit_change scenario

Environmental requirements: A road that has at least one change in the speed limit.

DUT behavior: The DUT drives on the road, presumably adapting its speed to the changing limitations.

13. DUT merge at lane end

13.1. Scenario description

Merging lanes requires the DUT to exercise both longitudinal and lateral control to make use of different sensors and decisional components. In this scenario, the DUT needs to merge from a lane that ends into a lane where another car is driving.

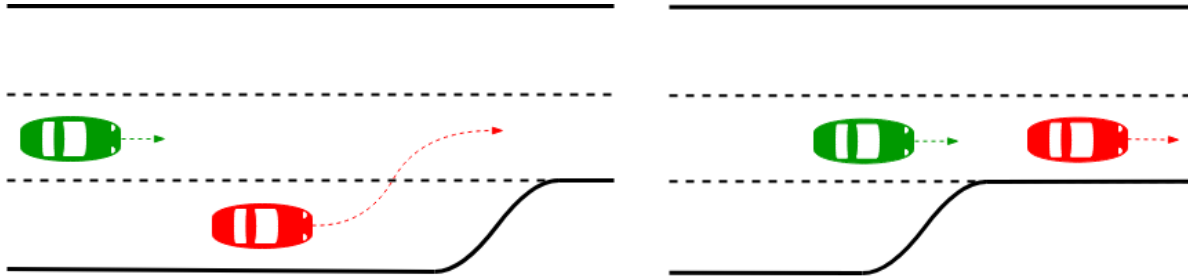


Figure 13-1 dut_merge_at_lane_end scenario

Environmental requirements: A road where a lane ends (in other words, with decreasing number of lanes).

DUT behavior: The DUT starts driving in the lane that ends and needs to merge into the neighboring lane.

Other actors' behavior: Another car drives in the neighboring lane, behind the DUT, at a distance that should allow the DUT to merge.

14. Scenarios with Stationary Objects

14.1. Stationary objects

Stationary objects such as puddles, traffic cones, pedestrians and so on can pose an additional challenge to the DUT in any scenario.

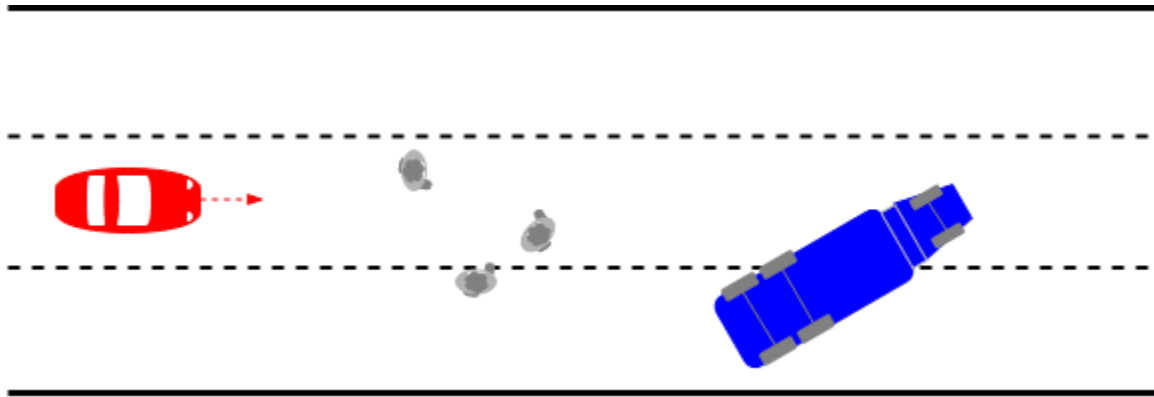


Figure 14-1 Pedestrians and truck on the path of the DUT

Environmental requirements: A road with at least one lane.

DUT behavior: The DUT starts driving in the lane that has stationary objects on it and takes an appropriate safe action.

Other actors' behavior: Objects are static, but pedestrians may move or cross the lane/road.